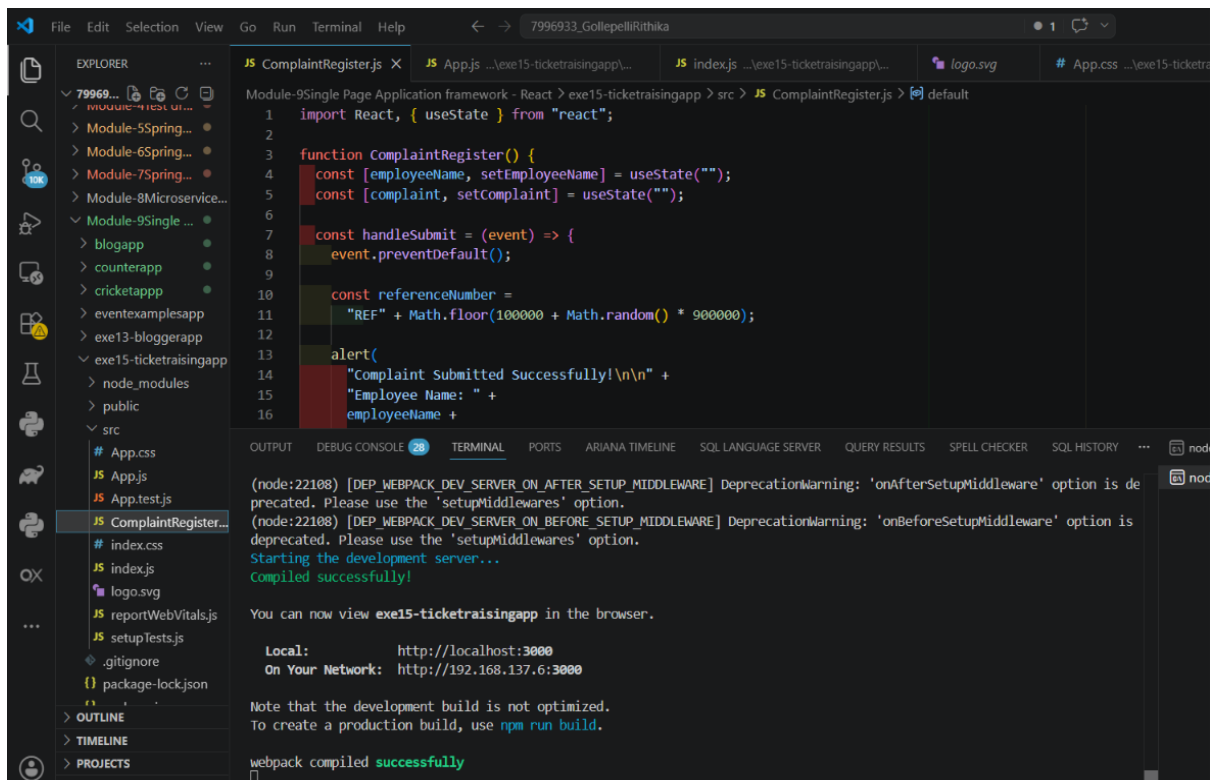


Output15exe15-ticketraisingapp.ReactJS-HOL



```
1 import React, { useState } from "react";
2
3 function ComplaintRegister() {
4   const [employeeName, setEmployeeName] = useState("");
5   const [complaint, setComplaint] = useState("");
6
7   const handleSubmit = (event) => {
8     event.preventDefault();
9
10    const referenceNumber =
11      "REF" + Math.floor(100000 + Math.random() * 900000);
12
13    alert(
14      "Complaint Submitted Successfully!\n\n" +
15      "Employee Name: " +
16      employeeName +
```

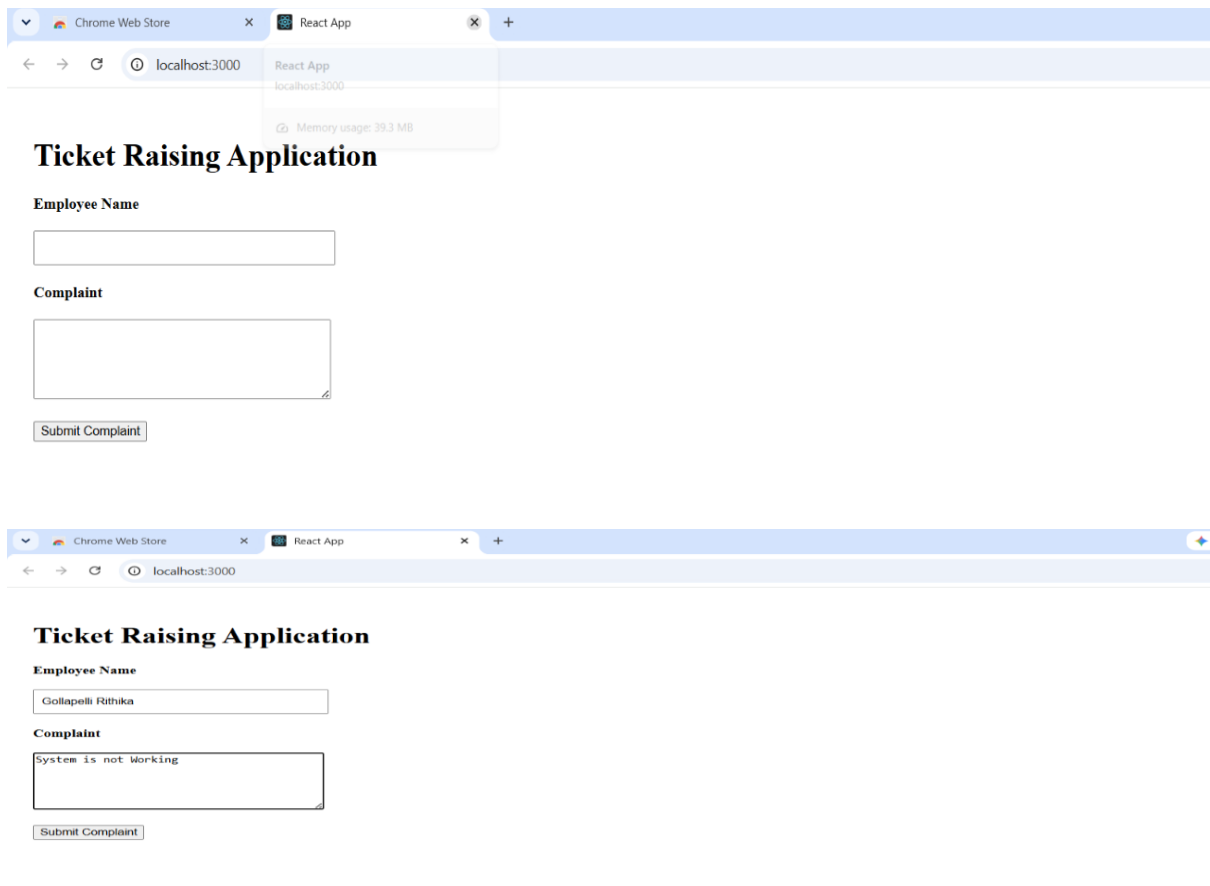
(node:22108) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:22108) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view exe15-ticketraisingapp in the browser.

Local: http://localhost:3000
On Your Network: http://192.168.137.6:3000

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled successfully



Ticket Raising Application

Employee Name

Complaint

Submit Complaint

Ticket Raising Application

Employee Name

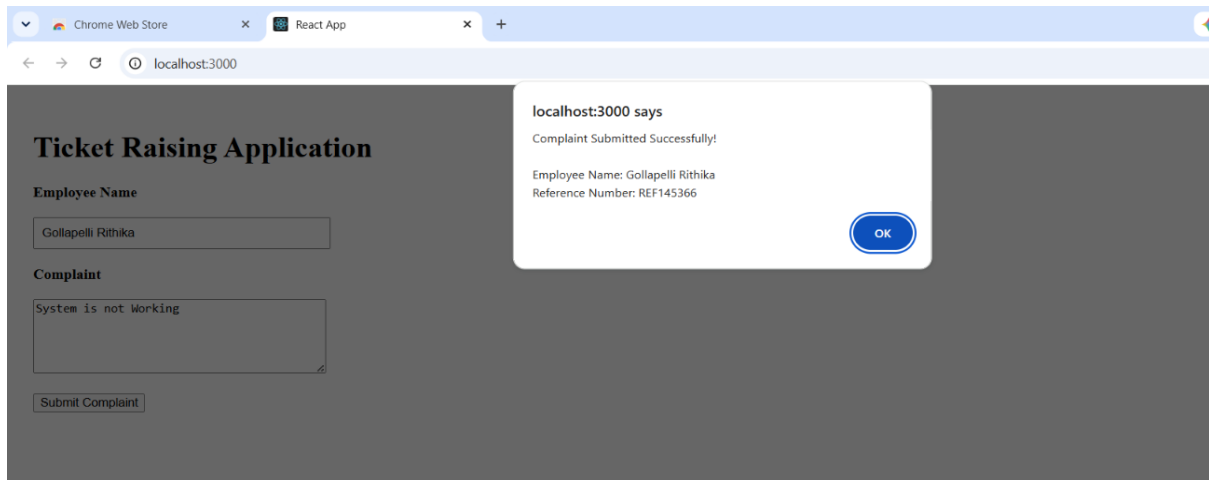
Complaint

Submit Complaint

After entering:

- Employee Name: Gollapelli Rithika
- Complaint: System is not working

and clicking Submit Complaint, an alert appears similar to:



The reference number will be different each time because it is generated randomly. This satisfies the HOL requirement to use a React form, input controls, handleSubmit, and generate a complaint reference number on submission.

Objectives

1. Explain About React Forms

React Forms are used to collect and manage user input, such as names, email addresses, passwords, feedback, or search queries. Unlike traditional HTML forms, React forms use state to control form data, making it easier to validate inputs, update values dynamically, and handle form submissions efficiently.

Benefits

- Manages user input efficiently.
- Provides real-time updates.
- Simplifies validation and error handling.
- Creates interactive and responsive forms.

2. Define Controlled Components

A Controlled Component is a form element whose value is controlled by React state. The displayed value always comes from the component's state, and any changes made by the user are handled through event handlers such as onChange.

Advantages

- Keeps form data synchronized with React state.
- Makes validation easier.
- Provides better control over user input.
- Simplifies debugging and testing.

3. Explain About Various Input Controls

React supports different HTML input controls to collect various types of user information.

Common Input Controls

Input Control Purpose

text	Accepts single-line text input.
password	Accepts hidden password input.
email	Accepts email addresses.
number	Accepts numeric values only.
radio	Allows selection of one option from multiple choices.
checkbox	Allows selection of multiple options.
select	Displays a dropdown list of options.
textarea	Accepts multi-line text input.
file	Allows users to upload files.
date	Allows users to select a date.

4. Explain About Handling Forms

Handling forms in React involves managing form data using state and updating it whenever the user interacts with input fields. The `onChange` event captures user input, while state stores the latest values.

Benefits

- Keeps form values synchronized with state.
- Makes validation and error handling easier.

5. Explain About Submitting Forms

Form submission in React is handled using the `onSubmit` event. The `event.preventDefault()` method is commonly used to stop the browser's default form submission behavior, allowing React to process the form data instead.

How It Works

- `onSubmit` is triggered when the form is submitted.
- `preventDefault()` prevents the page from reloading.
- The form data is processed using React state.
- The application can then validate, store, or send the data to a server.